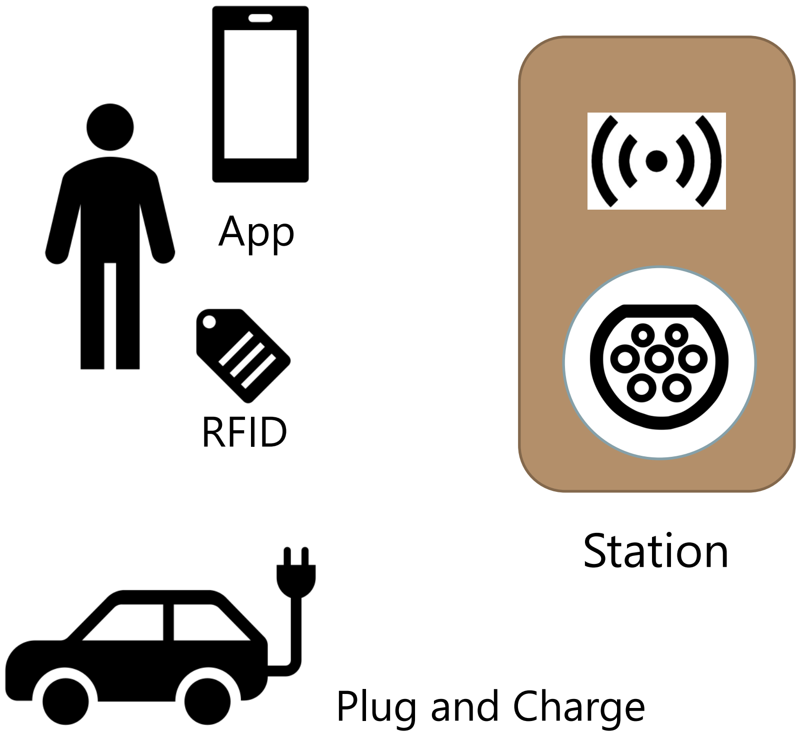
5. AUTHORIZE AND START CHARGING

Different ways to use stations

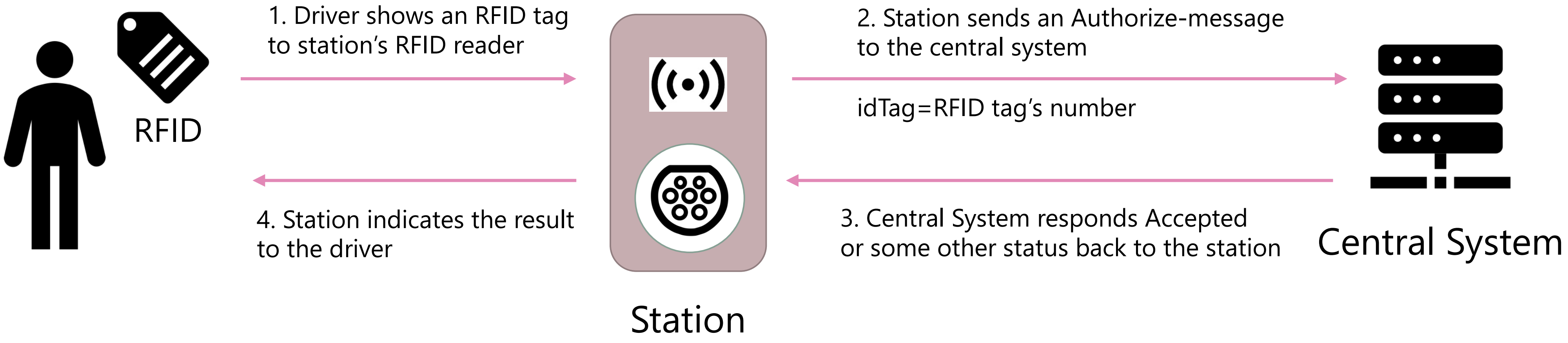


When a driver wants to use a charging station, the first step is often to **authorize** the driver. This is done to identify the driver and ensure that they are allowed to use the station.

There are **several ways** to authorize drivers:

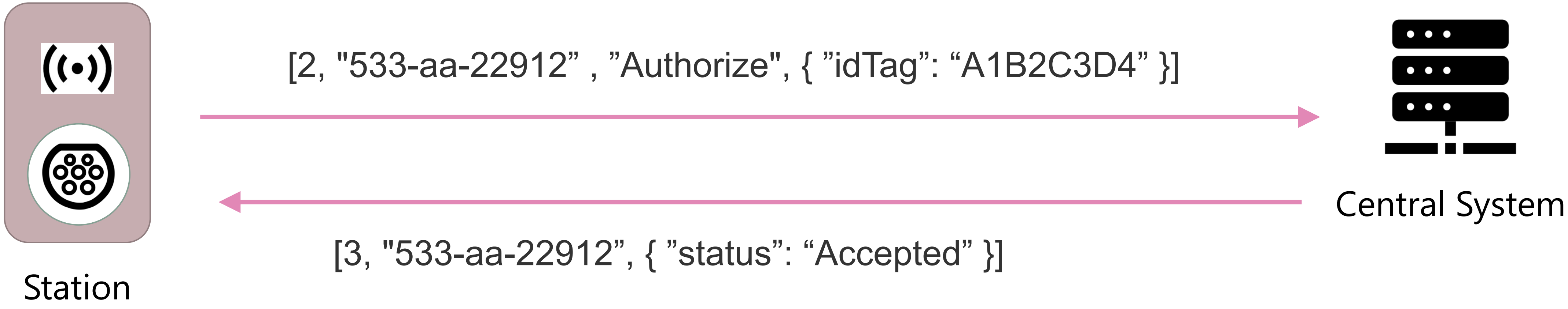
1. **RFID tags** – Drivers can show an RFID tags to the station's RFID reader to authorize themselves.
2. **Mobile apps** – Drivers can use mobile apps to authenticate themselves and use a station.
3. **Plug and Charge** – Drivers can also be authorized through the charging cable, provided that both the EV and the station support the ISO 15118 standard.

Basic Authorization Flow



The simplest authorization flow is through the use of **an RFID tag**. In this case, a driver shows the RFID tag to the station, which then sends an **Authorize** message with the RFID tag's number to the Central System. The Central System **maintains a list of accepted RFID tags** and checks to see if the provided RFID tag is in the list.

Example Authorize messages



In this simple example the station first sends **Authorize.req** message to the Central System.

The request message has one parameter, **idTag**, which has the RFID-tag’s unique number.

Usually this is sent in hexadecimal format.

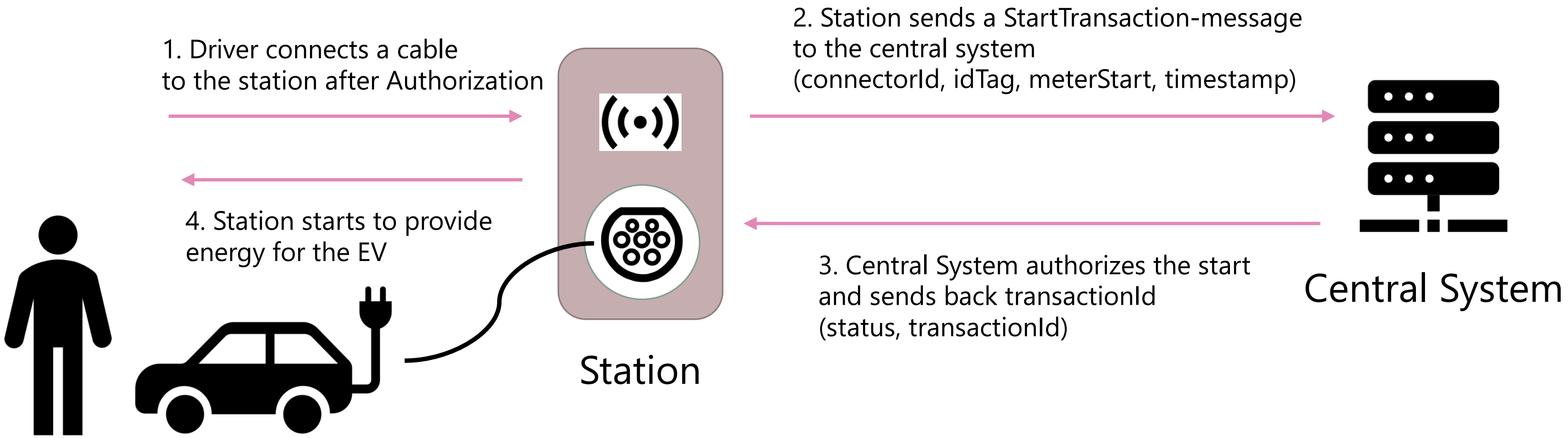
The Central System then sends a **Authorize.conf** with just one field **status**, where it tells whether it accepts or rejects the authorization attempt.

Authorization statuses

There are a few different **response statuses** that a central system can send back to the station in response to an Authorize request. **If the idTag is valid** and the user is allowed to charge, the central system will respond with an **Accepted** status. If the idTag is not valid or the user is not allowed to charge, the central system will give another status to tell why the idTag was rejected.

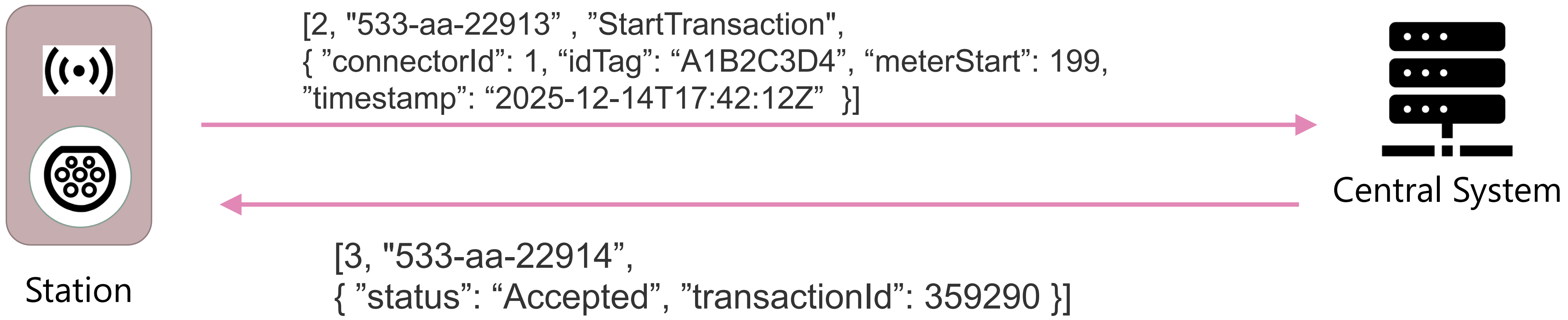
Status	Description
Accepted	IdTag is valid and can be used for charging.
Blocked	This indicates that the idTag has been blocked by the central system, either because it has been reported as lost or stolen, or because the user has been banned from using the charging network.
Expired	This indicates that the idTag has expired and can no longer be used for charging.
Invalid	This indicates that the idTag is not recognized by the central system, either because it has been typed incorrectly or because it is not associated with any known user account.

Basic Charging Flow



After a successful authorization using an RFID card, the station waits for a charging cable to be connected from the station to the EV. Once the station has detected that an EV is connected and ready, it sends a **StartTransaction** message to the central system to request permission to start charging. The central system then checks if the charging request is allowed or not, and if it is allowed, it returns **an unique transaction ID** to the station.

Example Start Transaction message



Once the EV is connected and ready to start charging, the station sends **StartTransaction.req** message to the Central System. The request has more detailed information of the charge that is now being started.

The Central System sends a **StartTransaction.conf** with **status** (usually accept or reject), and the Central system also generates a unique **transaction ID** for this charge.

StartTransaction.req fields

Field	Card.	Description
connectorId	1..1	Required, value > 0. This is the number of the connector where the EV is connected to and which is now being used for this charging transaction.
idTag	1..1	Required. In this simplest RFID tag use case, this is the unique RFID number of the tag. In other use cases, such as when mobile apps are used, the idTag can be something other than the RFID number. However, the idTag is always used to identify the person using the station and/or authorize whether using the station is allowed.
meterStart	1..1	Required. Meter value as Wh of the station's energy meter for the selected connector at the start of the charging transaction, before any energy is delivered to the EV.
timestamp	1..1	Required. Date and time when the charging transaction began. Please note that this may be different from the time when the OCPP message is delivered from the station to the central system.
reservationId	0..1	Optional. If reservations are used with the station, this indicates which reservation ended when this charging transaction began.

StartTransaction.conf fields

Field	Card.	Description
status	1..1	Required. The response from the central system indicating whether it accepts or rejects this transaction. The most common responses are "Accepted" or "Rejected," but there are also other possible status responses, such as "Blocked," "Expired," "Invalid," and "ConcurrentTx."
transactionId	1..1	Required. Unique ID created by the central system for this transaction. This transaction ID is later used in several places, such as delivering energy readings from the station to the central system or when the station reports that the transaction has ended.